NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA
THEORY EXAMINATION
Question Paper

Month and year of the Exam **December 2025** **Roll No.** 124104069
Program: **B. Tech**
Subject: **Network Synthesis & Filters** Semester: **3rd**
Maximum Marks: **50** Course No: **EEPC-205**
Number of Questions to be **fully attempted: Any 5** Time allowed.: **3 Hrs.**
Total No. of Questions: **6** Total No. of Pages Used: **2**

Unless stated otherwise, the symbols have their usual meanings in context with the subject. Assume suitably and state, additional data required, if any. The candidates before starting to write the solutions, should check the question paper for any discrepancies, and also ensure that they have delivered the question paper of the **right course no.** and the right subject title.

Q1 (a) Find the range of values of m in the following polynomial $P(s)$, so that the $P(s)$ is Hurwitz. 5

$$P(s) = 2s^4 + s^3 + ms^2 + s + 2$$

(b) Show that the following function is a positive real function. 5

$$F(s) = \frac{s^3 - 1}{(4s^3 - 3s^2 - 1)}$$

Q2 (a) Design a T section **constant-k** type high pass filter having a cut-off frequency of *12* kHz and a nominal characteristic impedance of **500 Ω.** Also find (i) its characteristic impedance and phase constant at 24 kHz and (ii) attenuation constant (α) at **4kHz** 5

(b) Design a **π-section** of **constant-k type** band pass filter having a lower and upper cut-off frequencies of **4 *kHz*** and ***10 kHz*** respectively and a nominal characteristic impedance of ***500 Ω.*** 5

Q3 (a) Define the terms: attenuation constant, propagation constant and phase constant along with their mathematical descriptions and significance. 5

(b) Let the open circuit impedance specifications are as follows: 5

$$z_{11}(s) = z_{22}(s) = \frac{s(s^2 + 16)(s^2 + 36)}{(s^2 + 1)(s^2 + 25)}$$

and

$$z_{12}(s) = z_{21}(s) = \frac{s(s^2 + 4)}{(s^2 + 1)(s^2 + 25)}$$

Draw the corresponding lattice network & state the range of possible values of **K** (variable **K** will come into the picture while applying the procedure).

Q4 (a) Realize the **Foster-II form** of the following admittance function 5

$$Y(s) = \frac{s(s^2+9)}{10(s^2+4)(s^2+25)}$$

(b) Realize the network using the basic synthesis procedure having the following impedance function: 5

$$Z(s) = \frac{s^2+2s+10}{s(s+5)}$$

Q5 (a) Design the π-section of m-derived low pass filter (LPF) having a design impedance (constant-K value) of $500\ \Omega$, cut-off frequency of $2200\ Hz$ and frequency of infinite attenuation (resonating frequency) of $f_\infty = 2500\ Hz$. 5

(b) An m-derived **HPF** has $f_c = 1700$ **Hz** and **m = 0.6**. Find the frequency of infinite attenuation (f_∞). 5

Q6 (a) What are the advantages, limitations & design steps of an active filter. 5

(b) How can we obtain a second order low pass filter from a bi-quadratic transfer function? Derive the transfer function for a low pass filter using an operational amplifier. 5

End of the question paper